

corrigé zakito

Énoncé

1

a

$$A = f^0(A) \subset B, \quad f(B) = \bigcup_{n \geq 0} f^{n+1}(A) \subset B.$$

b

$$A \subset C, \quad f(C) \subset C.$$

$$f^0(A) \subset C, \quad f^n(A) \subset C \Rightarrow f^{n+1}(A) \subset f(C) \subset C.$$

$$\forall n \in \mathbb{N}, \quad f^n(A) \subset C \Rightarrow B \subset C.$$

2

$$f^n(3) = 3^{2^n}, \quad i \mapsto -1 \mapsto 1 \mapsto 1.$$

$$B = \{3^{2^n} : n \in \mathbb{N}\} \cup \{i, -1, 1\}.$$

3

a

$$f \in C^0(\mathbb{R}_+), \quad 0 \leq x < y \Rightarrow f(y) - f(x) = \frac{(y-x)(x+y)}{4} > 0.$$

$$u_0 = 2, \quad v_0 = 4, \quad u_{n+1} = \frac{u_n^2 + 3}{4}, \quad v_{n+1} = \frac{v_n^2 + 3}{4}.$$

$$f^n(A) = [u_n, v_n] \Rightarrow f^{n+1}(A) = [f(u_n), f(v_n)] = [u_{n+1}, v_{n+1}].$$

$$\forall n \in \mathbb{N}, \quad f^n(A) = [u_n, v_n], \quad 0 \leq u_n \leq v_n.$$

b

$$f(x) - x = \frac{(x-1)(x-3)}{4}, \quad f(1) = 1, \quad f(3) = 3.$$

$$1 < u_n < 3 \Rightarrow 1 < u_{n+1} < u_n < 3.$$

$$u_n \downarrow \ell \in [1, 2], \quad \ell = f(\ell) \Rightarrow \ell \in \{1, 3\} \Rightarrow \ell = 1.$$

$$v_n > 3 \Rightarrow v_{n+1} > v_n > 3.$$

$$v_n \uparrow L < +\infty \Rightarrow L \geq 4, \quad L = f(L) \Rightarrow L \in \{1, 3\}, \quad \perp.$$

$$u_n \downarrow 1, \quad v_n \uparrow +\infty.$$

c

$$[u_n, v_n] \subset (1, +\infty) \Rightarrow B \subset (1, +\infty).$$

$$x > 1 \Rightarrow \exists N, M, \quad u_N < x, \quad v_M > x.$$

$$n = \max(N, M) \Rightarrow x \in [u_n, v_n] \subset B.$$

$$B = (1, +\infty).$$

4

$$X = \bigcup_{n \geq 0} f^n(E \setminus A) = (E \setminus A) \sqcup f(X), \quad f(X) \subset A.$$

$$E \setminus X \subset A, \quad A = f(X) \sqcup (E \setminus X).$$

$\varphi(X) = f(X) \subset A, \quad \varphi(E \setminus X) = E \setminus X \subset A.$
 $f|_X : X \longrightarrow f(X) \text{ bijective}, \quad \text{id}_{E \setminus X} : E \setminus X \longrightarrow E \setminus X \text{ bijective}.$
 $\varphi : E = X \sqcup (E \setminus X) \longrightarrow A = f(X) \sqcup (E \setminus X) \text{ bijective}.$

5

$u : E \hookrightarrow F, \quad v : F \hookrightarrow E, \quad A = v(F) \subset E.$
 $v \circ u : E \hookrightarrow A \xrightarrow{4} \exists b : E \xrightarrow{\sim} A.$
 $v : F \xrightarrow{\sim} A \Rightarrow v^{-1} \circ b : E \xrightarrow{\sim} F.$

6

$\Phi(A, B) = 2A \cup (2B + 1).$
 $n \in A \iff 2n \in \Phi(A, B), \quad n \in B \iff 2n + 1 \in \Phi(A, B).$
 $\Phi(A, B) = \Phi(C, D) \Rightarrow A = C, \quad B = D.$
 $\mathcal{P}(\mathbb{N})^2 \hookrightarrow \mathcal{P}(\mathbb{N}), \quad A \mapsto (A, \emptyset) : \mathcal{P}(\mathbb{N}) \hookrightarrow \mathcal{P}(\mathbb{N})^2.$
 $\mathcal{P}(\mathbb{N})^2 \simeq \mathcal{P}(\mathbb{N}).$

7

a

$\alpha(f) = \alpha(g), \quad n \in \mathbb{N} \Rightarrow \exists m \in \mathbb{N}, \quad p_{n+1}^{f(n)+1} = p_{m+1}^{g(m)+1}.$
 Unicité de la décomposition en facteurs premiers $\Rightarrow m = n, \quad f(n) = g(n).$
 $\alpha : \mathbb{N}^{\mathbb{N}} \hookrightarrow \mathcal{P}(\mathbb{N}).$

b

$A \mapsto \mathbf{1}_A : \mathcal{P}(\mathbb{N}) \hookrightarrow \mathbb{N}^{\mathbb{N}}.$
 $\mathbb{N}^{\mathbb{N}} \simeq \mathcal{P}(\mathbb{N}).$

8

a

$x = \sigma^0(x) \Rightarrow x \sim x.$
 $y = \sigma^k(x) \Rightarrow x = \sigma^{-k}(y) \Rightarrow y \sim x.$
 $y = \sigma^k(x), \quad z = \sigma^\ell(y) \Rightarrow z = \sigma^{k+\ell}(x) \Rightarrow x \sim z.$

b

$m = |E|, \quad x, \sigma(x), \dots, \sigma^m(x) \in E.$
 $\exists 0 \leq i < j \leq m, \quad \sigma^i(x) = \sigma^j(x) \Rightarrow \sigma^{j-i}(x) = x.$
 $\{k \geq 1 : \sigma^k(x) = x\} \neq \emptyset \Rightarrow n(x) = \min\{k \geq 1 : \sigma^k(x) = x\}.$

c

$0 \leq i < j < n(x), \quad \sigma^i(x) = \sigma^j(x) \Rightarrow \sigma^{j-i}(x) = x, \quad 0 < j - i < n(x), \quad \perp.$
 $k \in \mathbb{Z}, \quad k = qn(x) + r, \quad 0 \leq r < n(x) \Rightarrow \sigma^k(x) = \sigma^r(x).$
 $[x] = \{x, \sigma(x), \dots, \sigma^{n(x)-1}(x)\}, \quad |[x]| = n(x).$

9

a

$$x \in X_i \Rightarrow n(x) = n_i \Rightarrow \sigma_i^{n_i}(x) = \sigma^{n_i}(x) = x.$$

$$x \notin X_i \Rightarrow \sigma_i^{n_i}(x) = x.$$

$$\sigma_i^{n_i} = \text{id}_E, \quad \sigma_i^{-1} = \sigma_i^{n_i-1}.$$

b

$$i \neq j, \quad X_i \cap X_j = \emptyset, \quad \sigma(X_i) = X_i, \quad \sigma(X_j) = X_j.$$

$$\sigma_i \sigma_j(x) = \sigma_j \sigma_i(x) = \begin{cases} \sigma(x) & x \in X_i \cup X_j, \\ x & x \notin X_i \cup X_j. \end{cases}$$

c

$$N = \text{ppcm}(n_1, \dots, n_r), \quad E = \bigsqcup_{i=1}^r X_i.$$

$$x \in X_i \Rightarrow N = q_i n_i \Rightarrow \sigma^N(x) = (\sigma^{n_i})^{q_i}(x) = x.$$

$$\sigma^N = \text{id}_E.$$

10

a

$$E_k = f^k(E), \quad E_{k+1} = f^k(f(E)) \subset f^k(E) = E_k.$$

$$|E_0| \geq |E_1| \geq \dots \geq 1 \Rightarrow \exists p, \quad E_{p+1} = E_p.$$

$$p = \min\{k \in \mathbb{N} : E_{k+1} = E_k\} \Rightarrow \forall k < p, \quad E_{k+1} \subsetneq E_k.$$

$$E_{p+1} = E_p \Rightarrow \forall j \geq 0, \quad E_{p+j} = f^j(E_p) = E_p.$$

b

$$F = E_p, \quad f(F) = E_{p+1} = F \Rightarrow \sigma = f|_F : F \rightarrow F.$$

$$F \text{ fini} \Rightarrow \sigma \text{ bijective.}$$

$$k = \max(1, p) \Rightarrow \forall x \in E, \quad f^k(x) \in E_k = F.$$